MIL-T-36385C
29 September 1977
SUPERSEDING
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18 September 1975

MILITARY SPECIFICATION

TESTER, FLASH POINT, CLOSED TYPE, 115 VOLT, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1.1 SCOPE

1.1 This specification covers a Pensky-Martens closed-type flash point tester, suitable for determining the flash point of fuel oils.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-M-7298 - Manual, Technical: Commercial Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM D 93	Standard Method of Test for Flash Point by Pensky-Martens Closed Tester.
ASTM E 1	Standard Specifications for ASTM Thermometers
ASTM E 134	Standard Specifications for Pensky-Martens Closed Flash Tester.

(Application for copies should be addressed to the Amercian Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and as specified herein.

3.2 Design and construction. The design and construction of the closed-type flash point tester shall be in accordance with the requirements specified in ASTM E 134 for the Pensky-Martens closed-type flask point tester with an electric resistance-type heater. The tester shall consist of a test cup, a cover with operating mechanism, a flexible shaft for hand stirring, an electric motor for driving the stirring device, an electric heater, two thermometers, and a support stand.

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3.2.1 Test cup. The test cup shall be of brass or other suitable metal, and shall be in accordance with ASTM E 134.

3.2.2 Cover with operating mechanism. The cover with operating mechanism shall be equipped with a stirrer, thermometer well, and test flame burner. The cover shall provide even heat distribution through the cast-iron air bath and plated shield. The test cup cover mechanism shall incorporate a single spring-loaded knob control for opening and introducing the stainless-steel test flame burner into the aperture. The spring-loaded knob shall automatically close the aperture when released. A gas pilot burner shall be provided to maintain a needle valve-controlled test flame. The stirrer shall be equipped with two double-bladed propellers. A flexible shaft shall be furnished for hand stirring and shall have a coupling to fit the shaft of the stirrer. The themometer well shall be secured to the cover. A chrome-plated brass radiator shield shall be positioned in order to maintain the required distance between the cup and bath. Adequate rubber tubing connections shall be supplied. The cover and mechanism shall be in accordance with ASTM E 134.

3.2.3 Flexible shaft. The flexible shaft shall be furnished for hand stirring and shall have a coupling suitable to fit the shaft of the stirrer.

3.2.4 Electrical components.

3.2.4.1 Electric motor and stirring device. The electric motor shall operate on nominal 115 volts 60 Hz, AC. Motor shall be capable of driving the stirring mechanism at the rates specified in ASTM D 93.

3.2.4.2 Heater. The heater shall operate on nominal 115 volts 60 Hz, AC. Heater shall have a controller which is capable of varying the power from 0 to 750 watts.

* 3.2.5 Thermometers. Two thermometers conforming to the requirements of ASTM E1 shall be furnished. One thermometer shall have a low range of 20° to 230° Fahrenheit (F) in accordance with ASTM Thermometer Number 9F. The other thermometer shall have a high range of 200°F. to 700°F. in accordance with ASTM Thermometer Number 10F. The thermometers shall be permanently marked in accordance with ASTM E1.

3.2.6 Support stand. The stand shall be sufficiently sturdy to support the electric motor, heater, tubing, test cup, and holder for test cup.

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3.3 Performance. The flash point tester shall be capable of testing oils and suspensions of solids in accordance with ASTM D 93.

3.3.1 Oils. The tester shall produce the conditions specified in ASTM D 93 when tested as specified in 4.4.1.

3.3.2 Suspensions of solids. The tester shall produce the conditions specified in ASTM D 93 when tested as specified in 4.4.2.

3.4 Service data. The contractor shall furnish with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.5 Identification marking. Each tester shall be permanently and legibly marked with the following information: Manufacturer's name or trademark, contract or purchase order number, National Stock Number, letters "U.S.", the voltage, wattage and serial number. Markings secured by adhesive shall not be used.

3.6 Workmanship. Workmanship shall be first class throughout. The tester shall be free from defects cited in Table III and other defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the thermometers, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the Item Identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.3 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product for examination purposes shall be one tester.

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Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number shall be zero for all sample sizes. Unit of product shall be one tester.

Table II Sampling for tests

Component	Characteristic	Requirement	Test procedure	Inspection level
Tester	Performance with oil	3.3.1	4.4.1	S-1
Tester	Performance with suspensions of solids	3.3.2	4.4.2	S-1

4.3 Examination. The tester shall be examined to determine compliance with all requirements contained in this specification. The tester shall be examined for defects including, but not limited to, those listed in Table III.

Table III Classification of defects

Categories	Defects
Major	
101	Component part not present.
102	Unit not stable on flat, plane surface.
103	Metal parts not free of corrosion.
104	Component parts not free of sharp edges.
105	Shutter not operating properly.
106	Service data not complete, correct or legible.
107	Thermometers not marked correctly.
108	Cover with operating mechanism not complete.
Minor	
201	Control knobs not secure.
202	Cup not free of foreign matter.
203	Identification markings incorrect, incomplete or missing.

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4.3.1 Dimensional examination. The testers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall conducted to determine compliance with specification requirements.

4.4.1 Flash point of oils. Oil used shall have a flash point of not less than 650°F. Flash point shall be run in accordance with procedures specified in ASTM D 93 for fuel oils or lubricating oils. Stirring speed shall be measured with a stroboscope or tachometer.

4.4.2 Flash point of suspensions of solids. An oil which contains wax shall be used for test media. Flash point shall be run in accordance with procedures specified in ASTM D 93 for suspensions of solids. Stirring speed shall be measured with a stroboscope or tachometer.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE IV. Examination of preparation for delivery.

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each tester shall be packaged in a box of appropriate size and design and constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings, or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Each tester, packaged as specified in 5.1.1.1, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the box specification.

5.2.2 Level B. Each tester, packaged as specified in 5.1.1.1, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Units shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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(a) Title, number and date of this specification.
(b) Applicable National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6630-00-530-0987	TESTER, FLASH POINT, Closed Type, 115 volt, 60 Hz, AC

6.3 This specification does not cover all type, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-T-36385C TESTER, FLASH POINT, CLOSED TYPE, 115 V, 60 HZ, AC

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